RESULT 1 / NOTHING IS LOST

First, nothing is lost: RIGOR keeps the schema.

We climb from the easiest check to the hardest. Every number covers all 30 ontologies.

1 VALID | 2 KEEPS THE SCHEMA | 3 ADDS MEANING | 4 FEWER ERRORS | 5 GOLD MAPPINGS | 6 JUDGES

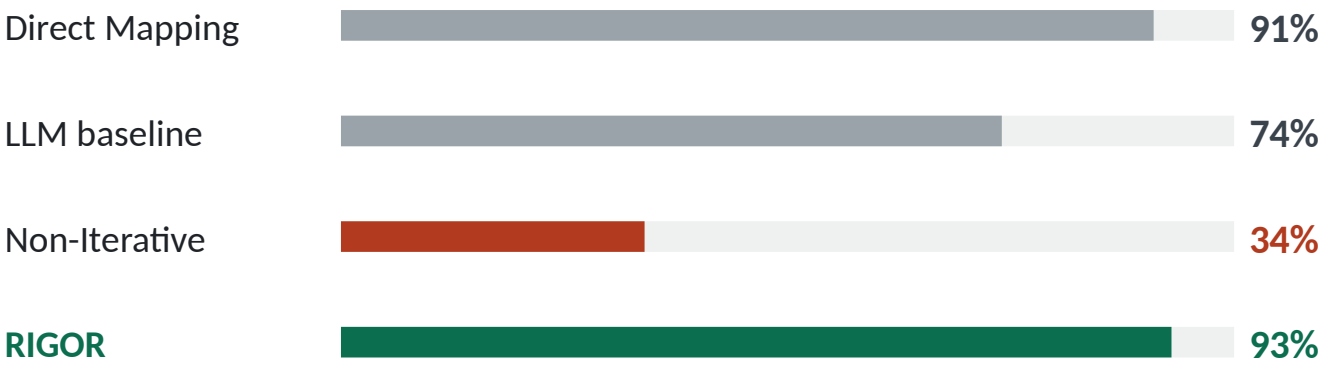
1 VALID

30 / 30

ontologies are valid OWL 2 DL and logically consistent, for every method.

2 KEEPS THE SCHEMA

share of database columns covered, per ontology



Same retrieved context, no seed: Non-Iterative covers only a third of the columns. RIGOR's direct-mapping seed keeps them.

Validity: RDFLib; consistency: HermiT. Coverage: a column counts as covered when an ontology class or property name is similar to it (embedding similarity ≥ 0.55), averaged over each method's ontologies. Paper Sec. 4.6–4.7, Table 9.

RESULT 2 / MEANING AND ERRORS

Then, meaning is added, with far fewer errors.

Averages per ontology, over all 30 ontologies.

1 VALID | 2 KEEPS THE SCHEMA | 3 ADDS MEANING | 4 FEWER ERRORS | 5 GOLD MAPPINGS | 6 JUDGES

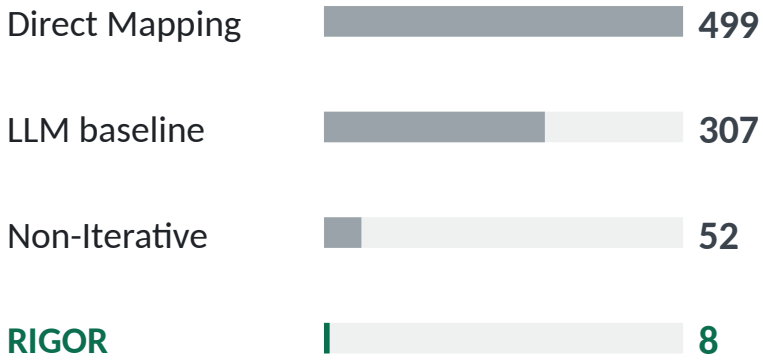
3 ADDS MEANING

elements per ontology

	Labels & comments	Provenance links	Subclass axioms
Direct Mapping	0	0	0
LLM baseline	0	0	0
Non-Iterative	**389**	0	0
RIGOR	**777**	**196**	**59**

4 FEWER ERRORS

OOPS! modeling pitfalls per ontology



72% of the others' pitfalls: missing annotations. The critical one, a second domain: **zero in 6 of 9** RIGOR ontologies.

Paper Table 8 (annotation assertions, prov:wasDerivedFrom, SubClassOf) and Table 7 (OOPS!; P08 missing annotations, P19 multiple domains or ranges).

RESULT 3 / EXTERNAL BENCHMARK

RIGOR keeps the link to the data.

BURR [Laskowski et al., PACMMOD 2025] · ISWC scenario · highest F1 among ontology-learning systems

1 VALID | 2 KEEPS THE SCHEMA | 3 ADDS MEANING | 4 FEWER ERRORS | 5 GOLD MAPPINGS | 6 JUDGES

WHAT BURR CHECKS

Every ontology element must map back to the database.

CLASSES	`person`	→	`Class: Person`
RELATIONS	`paper.author_id`	→	`ObjectProperty: author`
ATTRIBUTES	`person.name`	→	`DataProperty: name`

Elements match when they map to the same database elements, whatever their names. F1 per element type.

Illustrative elements · ISWC gold mapping: D2RQ example by Bizer & Seaborne

SYSTEM	CLASSES	RELATIONS	ATTRIBUTES
W3C-Mapper / Ontop rule-based	0.53	0.11	0.39
RDB2Onto rule-based	0.67	0.62	0.44
OntoGenix LLM-based	0.58	0.35	0.44
RIGOR LLM + rules	**0.73**	**0.67**	**0.52**

Relations: 0.67 vs 0.11 for W3C-Mapper / Ontop. Foreign keys become named relations, like hasPatientData.

One scenario · 9 tables · 46 columns · 11 FKs · RIGOR with Claude · its prov:wasDerivedFrom links exported as a D2RQ mapping · Source: paper Table 13

RESULT 4 / LLM JUDGE AND HUMAN EXPERTS

An LLM judge and human experts both rank RIGOR first.

Both ranked four anonymized ontologies per table, from first to fourth.

1 VALID | 2 KEEPS THE SCHEMA | 3 ADDS MEANING | 4 FEWER ERRORS | 5 GOLD MAPPINGS | 6 JUDGES

LLM-AS-A-JUDGE · GPT-5.4

58 / 58

tables with RIGOR ranked first

Liver Cancer 16/16 · eICU-CRD 31/31 · Chinook 11/11

Largest gains: clarity and domain enrichment.

10 ONTOLOGY EXPERTS

28 / 40

rankings with RIGOR first (70%)



Direct Mapping: structure without meaning. LLMs alone: meaning without structure. **RIGOR: both, and checked.**

Paper Table 1 (rankings), Table 10 (judge dimensions), Table 12 (expert agreement, Kendall's W 0.39–0.72). Judge: 290 competency questions. Experts: 4 medical tables.

RESULT 5 / WHERE THE GAINS COME FROM

RIGOR's gains do not depend on model size or memorized data.

Is it just a strong LLM? Is it memorized data? We broke the results down by LLM and by domain.

1 VALID | 2 KEEPS THE SCHEMA | 3 ADDS MEANING | 4 FEWER ERRORS | 5 GOLD MAPPINGS | 6 JUDGES

ACROSS LLMs · MISTRAL SMALL (24B)

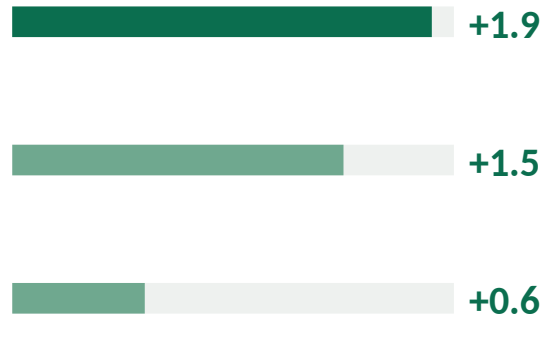
column coverage, average over the three databases



On par with Claude under RIGOR (**95%**), for **$0.12** instead of $33 for all three databases.

ACROSS DOMAINS · LLM JUDGE

RIGOR's lead over the best other method (score 0–5)



A small open model catches up, and the lead grows where the LLM knows least: **grounding and validation make the difference.**

Coverage: paper Table 9 (Claude 95%, DeepSeek 86% with RIGOR). Cost: Table 15. Judge scores: Table 1 (Claude ontologies; best other: Direct Mapping, Direct Mapping, Non-Iterative).